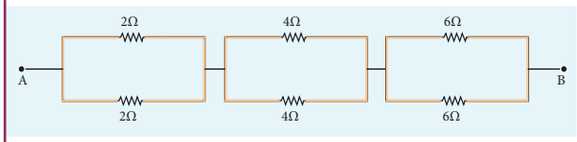
EXAMPLE 2.11

Calculate the equivalent resistance between A and B in the given circuit.



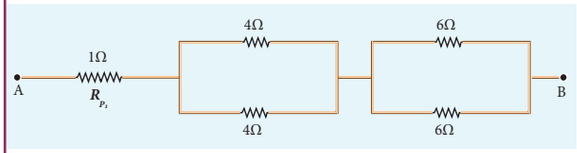
Solution

Parallel connection

Part 1

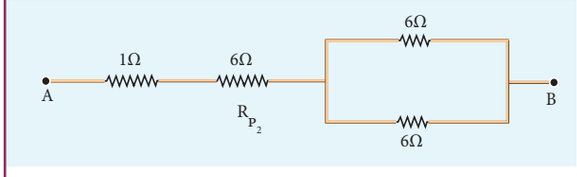
$$\frac{1}{R_{p_1}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{p_1}} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} \qquad R_{p_1} = 1\Omega$$



Part II

$$\frac{1}{R_{p_2}} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}, \quad \frac{1}{R_{p_2}} = \frac{1}{2}, \quad R_{p_2} = 2\Omega$$



Part III

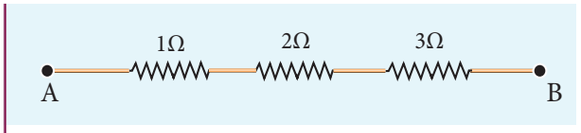
$$\frac{1}{R_{p_3}} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

$$\frac{1}{R_{p_3}} = \frac{1}{3}, \qquad R_{p_3} = 3\Omega$$

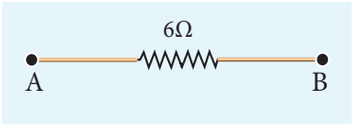
$$R = R_{p_1} + R_{p_2} + R_{p_3}$$

R = 1 + 2 + 3 R = 6 Ω

The circuit became:

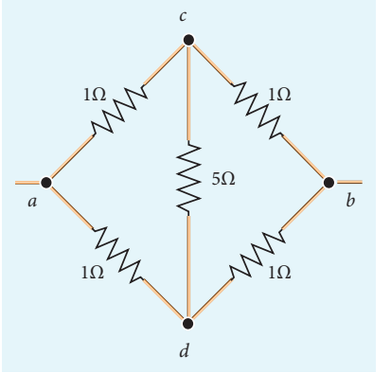


Equivalent resistance between A and B is



EXAMPLE 2.12

Five resistors are connected in the configuration as shown in the figure. Calculate the equivalent resistance between the points a and b.



Solution

Case (a)

To find the equivalent resistance between the points a and b, we assume that current is entering the junction a. Since all the resistances in the outside loop are the same (1Ω), the current in the branches ac and ad must be equal. So the electric potential at the point c and d is the same hence no current flows into 5 Ω resistance. It implies that the 5 Ω has no role in determining the equivalent resistance and it can be removed. So the circuit is simplified as shown in the figure.

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